

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**\_\_\_\_\_  
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1.  $\left[ \sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[ \sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2.  $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $(-1 - i)^3$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^3 + 8i = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z$ ,  $z^2 + z$ ,  $(z^2 + z)^2 + z$ ,  $[(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

|               |          |                |          |
|---------------|----------|----------------|----------|
| 5. a) $z = i$ | YES / NO | b) $z = 1 + i$ | YES / NO |
|               |          |                |          |

## ANSWER KEY – Non-Calculator

## SHOW ALL WORK FOR FULL CREDIT!

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Find the product. Write the product in rectangular form, using exact values. (3 points)

1.  $\left[ \sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[ \sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

## WORK

$$\begin{aligned} & \left[ \sqrt{6} \operatorname{cis} 165^\circ \right] \left[ \sqrt{2} \operatorname{cis} 255^\circ \right] \\ &= \sqrt{6} \cdot \sqrt{2} \operatorname{cis} (165^\circ + 255^\circ) \\ &= \sqrt{12} \operatorname{cis} 420^\circ \\ &= 2\sqrt{3} \operatorname{cis} 60^\circ \\ &= 2\sqrt{3} (\cos 60^\circ + i \sin 60^\circ) \\ &= 2\sqrt{3} \left( \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ &= \boxed{\sqrt{3} + 3i} \end{aligned}$$

Trap:  $165^\circ$  and  $255^\circ$  aren't unit-circle angles; only the sum is. Add first, then look.

## WHY

Rewrite each factor as  $r \operatorname{cis} \theta$ .

Product theorem: multiply the  $r$ 's, add the  $\theta$ 's.

$$\sqrt{12} = \sqrt{4}\sqrt{3} = 2\sqrt{3}. \quad 420^\circ - 360^\circ = 60^\circ.$$

Write cis out.

Unit circle at  $60^\circ$ .

$$\text{Distribute: } 2\sqrt{3} \cdot \frac{\sqrt{3}}{2} = 3.$$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2.  $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

## WORK

$$\begin{aligned} r_1 &= \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2 \\ \theta_1 &= \tan^{-1} \left( \frac{1}{-\sqrt{3}} \right) = -30^\circ \\ \theta_1 &= -30^\circ + 180^\circ = 150^\circ \\ r_2 &= \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{2+6} = 2\sqrt{2} \\ \theta_2 &= \tan^{-1} \left( \frac{-\sqrt{6}}{\sqrt{2}} \right) = \tan^{-1}(-\sqrt{3}) = -60^\circ \\ \theta_2 &= -60^\circ + 360^\circ = 300^\circ \\ \frac{2 \operatorname{cis} 150^\circ}{2\sqrt{2} \operatorname{cis} 300^\circ} &= \frac{2}{2\sqrt{2}} \operatorname{cis} (150^\circ - 300^\circ) \\ &= \frac{1}{\sqrt{2}} \operatorname{cis} (-150^\circ) \\ &= \frac{\sqrt{2}}{2} \operatorname{cis} 210^\circ \\ &= \frac{\sqrt{2}}{2} \left( -\frac{\sqrt{3}}{2} - \frac{1}{2} i \right) \\ &= \boxed{-\frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} i} \end{aligned}$$

## WHY

Top first:  $r = \sqrt{x^2 + y^2}$ .

$\tan^{-1}$  only gives QI or QIV.

$x < 0, y > 0$  is QII: add  $180^\circ$ .

Bottom.  $\sqrt{8} = 2\sqrt{2}$ .

$$\frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3}.$$

$x > 0, y < 0$  is QIV: add  $360^\circ$ .

Quotient theorem: divide the  $r$ 's, subtract the  $\theta$ 's.

Rationalize  $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ .  $-150^\circ + 360^\circ = 210^\circ$ .

$210^\circ$  is QIII: cos and sin both negative.

$$\sqrt{2} \cdot \sqrt{3} = \sqrt{6}.$$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $(-1 - i)^3$

WORK

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = 45^\circ$$

$$\theta = 45^\circ + 180^\circ = 225^\circ$$

$$(-1 - i)^3 = \left(\sqrt{2} \operatorname{cis} 225^\circ\right)^3$$

$$= (\sqrt{2})^3 \operatorname{cis}(3 \cdot 225^\circ)$$

$$= 2\sqrt{2} \operatorname{cis} 675^\circ$$

$$= 2\sqrt{2} \operatorname{cis} 315^\circ$$

$$= 2\sqrt{2} \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right)$$

$$= \boxed{2 - 2i}$$

WHY

Convert first.

*Trap:* both parts negative is QIII, not  $45^\circ$ .

De Moivre:  $r^n$  and  $n\theta$ .

$$(\sqrt{2})^3 = \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} = 2\sqrt{2}.$$

$$675^\circ - 360^\circ = 315^\circ.$$

$315^\circ$  is QIV:  $\cos +$ ,  $\sin -$ .

$$2\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 2.$$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^3 + 8i = 0$

WORK

$$x^3 = -8i$$

$$r = \sqrt{0^2 + (-8)^2} = 8$$

$$\theta = 270^\circ$$

$$x = \sqrt[3]{8} \operatorname{cis} \left(\frac{270^\circ + 360^\circ k}{3}\right)$$

$$= 2 \operatorname{cis}(90^\circ + 120^\circ k)$$

$$k = 0: 2 \operatorname{cis} 90^\circ = 2(0 + 1i) = 2i$$

$$k = 1: 2 \operatorname{cis} 210^\circ = 2\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\sqrt{3} - i$$

$$k = 2: 2 \operatorname{cis} 330^\circ = 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \sqrt{3} - i$$

$$\boxed{x = 2i, -\sqrt{3} - i, \sqrt{3} - i}$$

*Trap:*  $-8i$  is  $270^\circ$ , not  $90^\circ$ . Three roots for a cube, then stop.

WHY

Isolate  $x^3$ :  $+8i$  crosses over as  $-8i$ .

Convert  $0 - 8i$ .

Straight down the imaginary axis ( $x = 0$ , so no  $\tan^{-1}$ ).

$n$ th root theorem,  $k = 0, 1, 2$ .

$$\sqrt[3]{8} = 2. \text{ Cube roots sit } 120^\circ \text{ apart.}$$

QIII: both negative.

QIV:  $\cos +$ ,  $\sin -$ .

$$\text{Check: } (2i)^3 = 8i^3 = -8i.$$

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z$ ,  $z^2 + z$ ,  $(z^2 + z)^2 + z$ ,  $[(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

| 5. a) $z = i$                                                                                                                                                                                                           | YES / NO | b) $z = 1 + i$                                                                                                                                                                                                                                                                               | YES / NO |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>WORK</b><br>a) $z = i$ : $ i  = 1$<br>$z^2 + z = i^2 + i = -1 + i$<br>$ -1 + i  = \sqrt{1+1} = \sqrt{2} \approx 1.4$<br>$(-1 + i)^2 + i = (1 - 2i + i^2) + i = -i$<br>$ -i  = 1$<br>$(-i)^2 + i = -1 + i$ <b>YES</b> |          | <b>WHY</b><br>$ a + bi  = \sqrt{a^2 + b^2}$ . Not over 2.<br>$i^2 = -1$ .<br>Not over 2, keep going.<br>Square the last result, add $z$ .<br><br>It repeats $(-1 + i, -i, \dots)$ and never passes 2.<br>Not over 2.<br><br><i>Trap:</i> $(1 + i)^2 = 2i$ , not $1 + i^2$ .<br>Over 2: stop. |          |
| b) $z = 1 + i$ : $ 1 + i  = \sqrt{2} \approx 1.4$<br>$z^2 + z = (1 + i)^2 + (1 + i)$<br>$= (1 + 2i + i^2) + 1 + i = 1 + 3i$<br>$ 1 + 3i  = \sqrt{1+9} = \sqrt{10} \approx 3.2$ <b>NO</b>                                |          |                                                                                                                                                                                                                                                                                              |          |

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Find the product. Write the product in rectangular form, using exact values. (3 points)

1.  $\left[ \sqrt{2} \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[ \sqrt{2} \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

Find the quotient. Write it in rectangular form, using exact values. (3 points)

2.  $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $\left[ \sqrt{2} \left( \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^4 + 8 + 8i\sqrt{3} = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z$ ,  $z^2 + z$ ,  $(z^2 + z)^2 + z$ ,  $[(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a)  $z = -2$

YES / NO

b)  $z = -1 + i$

YES / NO

## ANSWER KEY – Non-Calculator

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Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Find the product. Write the product in rectangular form, using exact values. (3 points)

$$1. \left[ \sqrt{2} \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[ \sqrt{2} \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$$

## WORK

$$\begin{aligned} &= \sqrt{2} \cdot \sqrt{2} \operatorname{cis} \left( \frac{5\pi}{4} + \frac{11\pi}{12} \right) \\ &= 2 \operatorname{cis} \left( \frac{15\pi}{12} + \frac{11\pi}{12} \right) \\ &= 2 \operatorname{cis} \frac{26\pi}{12} = 2 \operatorname{cis} \frac{13\pi}{6} \\ &= 2 \operatorname{cis} \frac{\pi}{6} \\ &= 2 \left( \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \\ &= 2 \left( \frac{\sqrt{3}}{2} + \frac{1}{2}i \right) \\ &= \boxed{\sqrt{3} + i} \end{aligned}$$

Trap: given in radians, stay in radians.

## WHY

Product theorem: multiply the  $r$ 's, add the  $\theta$ 's.

Common denominator:  $\frac{5\pi}{4} = \frac{15\pi}{12}$ .

Simplify.

$\frac{13\pi}{6} - \frac{12\pi}{6} = \frac{\pi}{6}$  (subtract  $2\pi$ ).

Write cis out.

Unit circle at  $\frac{\pi}{6}$ .

Find the quotient. Write it in rectangular form, using exact values. (3 points)

$$2. \frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$$

## WORK

$$r_1 = \sqrt{(\sqrt{2})^2 + (\sqrt{6})^2} = \sqrt{2+6} = 2\sqrt{2}$$

$$\theta_1 = \tan^{-1} \left( \frac{\sqrt{6}}{\sqrt{2}} \right) = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$r_2 = \sqrt{(-\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2$$

$$\theta_2 = \tan^{-1} \left( \frac{-1}{-\sqrt{3}} \right) = \tan^{-1} \left( \frac{1}{\sqrt{3}} \right) = 30^\circ$$

$$\theta_2 = 30^\circ + 180^\circ = 210^\circ$$

$$\frac{2\sqrt{2} \operatorname{cis} 60^\circ}{2 \operatorname{cis} 210^\circ} = \frac{2\sqrt{2}}{2} \operatorname{cis} (60^\circ - 210^\circ)$$

$$= \sqrt{2} \operatorname{cis} (-150^\circ)$$

$$= \sqrt{2} \operatorname{cis} 210^\circ$$

$$= \sqrt{2} \left( -\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

$$= \boxed{-\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i}$$

## WHY

Top first.  $\sqrt{8} = 2\sqrt{2}$ .

Both positive: QI, no fix.

Bottom.

Trap: both negative is QIII, not  $30^\circ$ .

Quotient theorem: divide the  $r$ 's, subtract the  $\theta$ 's.

$-150^\circ + 360^\circ = 210^\circ$ .

QIII: both negative.

$\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$ .

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $\left[ \sqrt{2} \left( \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

WORK

$$\begin{aligned} &= (\sqrt{2})^4 \operatorname{cis} \left( 4 \cdot \frac{2\pi}{3} \right) \\ &= 4 \operatorname{cis} \frac{8\pi}{3} \\ &= 4 \operatorname{cis} \frac{2\pi}{3} \\ &= 4 \left( -\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \\ &= \boxed{-2 + 2\sqrt{3}i} \end{aligned}$$

WHY

De Moivre:  $r^n$  and  $n\theta$ .

$$(\sqrt{2})^4 = (\sqrt{2})^2(\sqrt{2})^2 = 4, \text{ not } 4\sqrt{2}.$$

$$\frac{8\pi}{3} - \frac{6\pi}{3} = \frac{2\pi}{3} \text{ (subtract } 2\pi).$$

$$\frac{2\pi}{3} \text{ is QII: } \cos -, \sin +.$$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^4 + 8 + 8i\sqrt{3} = 0$

WORK

$$\begin{aligned} x^4 &= -8 - 8i\sqrt{3} \\ r &= \sqrt{(-8)^2 + (-8\sqrt{3})^2} = \sqrt{64 + 192} = 16 \\ \theta &= \tan^{-1} \left( \frac{-8\sqrt{3}}{-8} \right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \\ \theta &= \frac{\pi}{3} + \pi = \frac{4\pi}{3} \\ x &= \sqrt[4]{16} \operatorname{cis} \left( \frac{4\pi/3 + 2\pi k}{4} \right) \\ &= 2 \operatorname{cis} \left( \frac{\pi}{3} + \frac{\pi}{2}k \right) \\ k=0: \quad 2 \operatorname{cis} \frac{\pi}{3} &= 2 \left( \frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = 1 + \sqrt{3}i \\ k=1: \quad 2 \operatorname{cis} \frac{5\pi}{6} &= 2 \left( -\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = -\sqrt{3} + i \\ k=2: \quad 2 \operatorname{cis} \frac{4\pi}{3} &= 2 \left( -\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = -1 - \sqrt{3}i \\ k=3: \quad 2 \operatorname{cis} \frac{11\pi}{6} &= 2 \left( \frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \sqrt{3} - i \\ x &= \boxed{1 + \sqrt{3}i, \quad -\sqrt{3} + i, \quad -1 - \sqrt{3}i, \quad \sqrt{3} - i} \end{aligned}$$

WHY

Isolate  $x^4$ : both terms flip sign.

$$(8\sqrt{3})^2 = 64 \cdot 3 = 192. \quad \sqrt{256} = 16.$$

*Trap:* both negative is QIII: add  $\pi$ .

$n$ th root theorem,  $k = 0, 1, 2, 3$ .

$$\sqrt[4]{16} = 2. \text{ Fourth roots sit } \frac{2\pi}{4} = \frac{\pi}{2} \text{ apart.}$$

$$\frac{\pi}{3} + \frac{\pi}{2} = \frac{2\pi}{6} + \frac{3\pi}{6} = \frac{5\pi}{6}.$$

$$\text{Check: } (2 \operatorname{cis} \frac{\pi}{3})^4 = 16 \operatorname{cis} \frac{4\pi}{3}.$$

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z$ ,  $z^2 + z$ ,  $(z^2 + z)^2 + z$ ,  $[(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>5. a) <math>z = -2</math></b> <span style="float: right;">YES / NO</span></p> <p><b>WORK</b></p> <p>a) <math>z = -2</math>: <math> -2  = 2</math></p> <p><math>z^2 + z = (-2)^2 + (-2) = 4 - 2 = 2</math></p> <p><math> 2  = 2</math></p> <p><math>(2)^2 + (-2) = 2</math> <span style="border: 1px solid black; padding: 2px;">YES</span></p> <p>b) <math>z = -1 + i</math>: <math> -1 + i  = \sqrt{2} \approx 1.4</math></p> <p><math>z^2 + z = (-1 + i)^2 + (-1 + i)</math></p> <p style="padding-left: 40px;"><math>= (1 - 2i + i^2) - 1 + i = -1 - i</math></p> <p><math> -1 - i  = \sqrt{2} \approx 1.4</math></p> <p><math>(-1 - i)^2 + (-1 + i) = (1 + 2i + i^2) - 1 + i</math></p> <p style="padding-left: 40px;"><math>= 2i - 1 + i = -1 + 3i</math></p> <p><math> -1 + 3i  = \sqrt{1 + 9} = \sqrt{10} \approx 3.2</math> <span style="border: 1px solid black; padding: 2px;">NO</span></p> | <p><b>b) <math>z = -1 + i</math></b> <span style="float: right;">YES / NO</span></p> <p><b>WHY</b></p> <p>Equal to 2 doesn't exceed 2: keep going.</p><br><br><p>Stuck at 2 forever, never over 2.</p> <p>Not over 2.</p><br><br><p><math>(-1 + i)^2 = 1 - 2i - 1 = -2i</math>.</p> <p>Still not over 2.</p> <p>Square the last result, add <math>z</math>.</p><br><br><p><i>Trap: it fails on the third check. Don't stop early and call it YES.</i></p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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Find the product. Write the product in rectangular form, using exact values. (3 points)

1.  $\left[ 2\sqrt{3} \left( \cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[ \sqrt{3} \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2.  $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $\left[ \sqrt{2} \left( \cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^4 + 4 = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

|                |          |             |          |
|----------------|----------|-------------|----------|
| 5. a) $z = -i$ | YES / NO | b) $z = 2i$ | YES / NO |
|                |          |             |          |

## ANSWER KEY – Non-Calculator

## SHOW ALL WORK FOR FULL CREDIT!

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Find the product. Write the product in rectangular form, using exact values. (3 points)

$$1. \left[ 2\sqrt{3} \left( \cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[ \sqrt{3} \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$$

## WORK

$$\begin{aligned} &= 2\sqrt{3} \cdot \sqrt{3} \operatorname{cis} \left( \frac{13\pi}{12} + \frac{5\pi}{4} \right) \\ &= 6 \operatorname{cis} \left( \frac{13\pi}{12} + \frac{15\pi}{12} \right) \\ &= 6 \operatorname{cis} \frac{28\pi}{12} = 6 \operatorname{cis} \frac{7\pi}{3} \\ &= 6 \operatorname{cis} \frac{\pi}{3} \\ &= 6 \left( \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ &= \boxed{3 + 3\sqrt{3}i} \end{aligned}$$

Trap:  $\frac{13\pi}{12}$  isn't on the unit circle, but the sum is. Add first, then look.

## WHY

Product theorem: multiply the  $r$ 's, add the  $\theta$ 's.

$$2\sqrt{3} \cdot \sqrt{3} = 2 \cdot 3 = 6. \quad \frac{5\pi}{4} = \frac{15\pi}{12}.$$

Simplify.

$$\frac{7\pi}{3} - \frac{6\pi}{3} = \frac{\pi}{3} \text{ (subtract } 2\pi\text{)}.$$

Unit circle at  $\frac{\pi}{3}$ .

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

$$2. \frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$$

## WORK

$$r_1 = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$$\theta_1 = \tan^{-1} \left( \frac{-2\sqrt{3}}{-2} \right) = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$\theta_1 = 60^\circ + 180^\circ = 240^\circ$$

$$r_2 = \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{2 + 6} = 2\sqrt{2}$$

$$\theta_2 = \tan^{-1} \left( \frac{-\sqrt{6}}{\sqrt{2}} \right) = \tan^{-1}(-\sqrt{3}) = -60^\circ$$

$$\theta_2 = -60^\circ + 360^\circ = 300^\circ$$

$$\frac{4 \operatorname{cis} 240^\circ}{2\sqrt{2} \operatorname{cis} 300^\circ} = \frac{4}{2\sqrt{2}} \operatorname{cis} (240^\circ - 300^\circ)$$

$$= \frac{2}{\sqrt{2}} \operatorname{cis}(-60^\circ)$$

$$= \sqrt{2} \operatorname{cis} 300^\circ$$

$$= \sqrt{2} \left( \frac{1}{2} - \frac{\sqrt{3}}{2} i \right)$$

$$= \boxed{\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2} i}$$

## WHY

Top first.

Trap: both negative is QIII, not  $60^\circ$ .

Bottom.

QIV: add  $360^\circ$ .

Quotient theorem: divide the  $r$ 's, subtract the  $\theta$ 's.

$$\frac{2}{\sqrt{2}} = \sqrt{2}. \quad -60^\circ + 360^\circ = 300^\circ.$$

QIV: cos +, sin -.

$$\sqrt{2} \cdot \sqrt{3} = \sqrt{6}.$$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $\left[ \sqrt{2} \left( \cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

WORK

$$\begin{aligned} &= (\sqrt{2})^4 \operatorname{cis} \left( 4 \cdot \frac{7\pi}{12} \right) \\ &= 4 \operatorname{cis} \frac{28\pi}{12} = 4 \operatorname{cis} \frac{7\pi}{3} \\ &= 4 \operatorname{cis} \frac{\pi}{3} \\ &= 4 \left( \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ &= \boxed{2 + 2\sqrt{3}i} \end{aligned}$$

Trap:  $\frac{7\pi}{12}$  isn't a unit-circle angle, but  $4 \cdot \frac{7\pi}{12}$  reduces to one. Multiply first, then reduce.

WHY

De Moivre:  $r^n$  and  $n\theta$ .

$(\sqrt{2})^4 = 4$ . Simplify the angle.

$\frac{7\pi}{3} - \frac{6\pi}{3} = \frac{\pi}{3}$  (subtract  $2\pi$ ).

Unit circle at  $\frac{\pi}{3}$ .

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^4 + 4 = 0$

WORK

$$x^4 = -4$$

$$r = 4, \quad \theta = 180^\circ$$

$$x = \sqrt[4]{4} \operatorname{cis} \left( \frac{180^\circ + 360^\circ k}{4} \right)$$

$$= \sqrt{2} \operatorname{cis} (45^\circ + 90^\circ k)$$

$$k = 0: \sqrt{2} \operatorname{cis} 45^\circ = \sqrt{2} \left( \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \right) = 1 + i$$

$$k = 1: \sqrt{2} \operatorname{cis} 135^\circ = \sqrt{2} \left( -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \right) = -1 + i$$

$$k = 2: \sqrt{2} \operatorname{cis} 225^\circ = \sqrt{2} \left( -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i \right) = -1 - i$$

$$k = 3: \sqrt{2} \operatorname{cis} 315^\circ = \sqrt{2} \left( \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i \right) = 1 - i$$

$$\boxed{x = 1 + i, -1 + i, -1 - i, 1 - i}$$

Trap:  $x = \pm\sqrt{2}i$  is wrong:  $(\sqrt{2}i)^4 = 4$ , not  $-4$ . The roots sit between the axes.

WHY

Isolate  $x^4$ .

$-4 + 0i$  sits on the negative real axis.

$n$ th root theorem,  $k = 0, 1, 2, 3$ .

$\sqrt[4]{4} = \sqrt{2}$ . Fourth roots sit  $90^\circ$  apart.

$\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 1$ .

QII.

QIII.

QIV.

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z$ ,  $z^2 + z$ ,  $(z^2 + z)^2 + z$ ,  $[(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a)  $z = -i$

YES / NO

WORK

a)  $z = -i: | -i | = 1$

$$z^2 + z = (-i)^2 - i = -1 - i$$

$$| -1 - i | = \sqrt{2} \approx 1.4$$

$$(-1 - i)^2 - i = (1 + 2i + i^2) - i = i$$

$$|i| = 1$$

$$i^2 - i = -1 - i \quad \boxed{\text{YES}}$$

b)  $z = 2i: |2i| = 2$

$$z^2 + z = (2i)^2 + 2i = -4 + 2i$$

$$| -4 + 2i | = \sqrt{16 + 4} = \sqrt{20} \approx 4.5 \quad \boxed{\text{NO}}$$

b)  $z = 2i$

YES / NO

WHY

Not over 2.

$$(-i)^2 = i^2 = -1.$$

Not over 2, keep going.

Square the last result, add  $z$ .

It repeats  $(-1 - i, i, \dots)$  and never passes 2.

Exactly 2 is not over 2: keep going.

$$(2i)^2 = 4i^2 = -4.$$

Over 2: stop.